

AD 502 – Machine Learning

Pre-Reading Material – Lecture 17

Naïve Bayes

1. What is Naive Bayes?

Suppose you receive an email saying:

“Congratulations! You have won a free prize.”

How can a machine decide whether this email is **Spam** or **Not Spam**?

One approach is to look at the words and other features in the email and estimate which class is more likely.

Naive Bayes is a classification technique that uses **probability** and **Bayes' theorem** to make this kind of decision.

It is not one single algorithm, but a family of related classification algorithms.

2. A Little Probability

You may have already encountered two simple ideas:

Conditional probability means finding the probability of something when we already know some other information.

For example:

What is the probability that a student passes **given that** the student studied?

Joint probability considers two events happening together.

You do not need to revise the mathematics deeply for this lecture. Just remember:

Probability helps us measure how likely something is to happen.

3. What is Bayes' Theorem?

Sometimes we know the probability of one situation, but we want to find the probability in the **opposite direction**.

For example:

If spam emails often contain the word “free”, can we use the presence of “free” to decide whether an email is spam?

Bayes' theorem provides a way to update our belief about a class after observing some evidence.

In simple words:

Start with what you already know, look at the new evidence, and update your belief.

4. Why is it called "Naive"?

Suppose we are classifying an email using features such as:

- "free" appears
- "offer" appears
- "winner" appears

Naive Bayes makes a **simplifying assumption**:

It treats the features as if they are independent of each other.

In real life, features may not actually be independent. That's why the assumption is called **naive**.

Even with this simple assumption, Naive Bayes can be surprisingly useful.

5. Where is Naive Bayes Used?

Naive Bayes is commonly used for classification tasks such as:

- **Spam email detection**
- **Document classification**
- **News classification**
- **Text classification**

For example, a news article might be classified as:

Sports | Politics | Technology | Entertainment

6. Different Types

You will encounter three important versions:

Multinomial Naive Bayes

Useful for text and document classification, where word frequencies are important.

Bernoulli Naive Bayes

Useful when features are represented as yes/no or present/absent.

Gaussian Naive Bayes

Useful when features are continuous numerical values.

7. One Important Problem

Naive Bayes can sometimes encounter a **zero-frequency problem**.

For example, if a particular word or feature was never observed in a class during training, the model may assign it zero probability.

A technique called **Laplace smoothing** can be used to handle this problem.

Think Before the Lecture

Don't worry about solving anything yet. Just think about these questions:

1. How could a computer identify a spam email?
2. Why might the words in an email provide useful information?
3. Why might assuming that all features are independent be unrealistic?
4. If a word never appeared in the training data, could that create a problem?

Key Idea

Naive Bayes uses probability to decide which class is most likely for a given set of features.

We will work through the actual calculations and examples in the lecture.

